Practical Session 2: Stability and Feedback

Control of Aerospace Systems

Unai Arce 100549832

August 10, 2026

1 Introduction

This report analyzes the lateral position control system for a navy aircraft landing on a carrier. The system consists of a pilot, modeled as a Proportional Integral (PI) controller, and the aircraft's physical dynamics. This PI action is evident in the controller's transfer function, which introduces a pole at the origin and a zero. The goal is to ensure stability while maintaining the centerline approach.

The open loop transfer function is:

$$G(s) = \frac{K(s+1)}{s(s-1)(s^2+10s+40)} = \frac{K(s+1)}{s^4+9s^3+30s^2-40s} \tag{1}$$

2 Task 1: Stability Analysis (Routh-Hurwitz)

The characteristic equation is $s^4 + 9s^3 + 30s^2 + (K-40)s + K = 0$.

s^4	1	30	K
s^3	9	$K-40$	0
s^2	$\frac{310-K}{9}$	K	0
s^1	$\frac{-K^2+269K-12400}{310-K}$	0	0
s^0	K	0	0

Solving the inequalities for no sign changes in the first column, we find that $K < 310$ from the s^2 row, $59.07 < K < 209.93$ from the s^1 row and $k > 0$ from the last row. Therefore we find the stability range:

$$\mathbf{59.07 < K < 209.93} \tag{2}$$

3 Task 2: Frequency Response Analysis

We evaluate the system's stability using Bode diagrams for five different values of K.

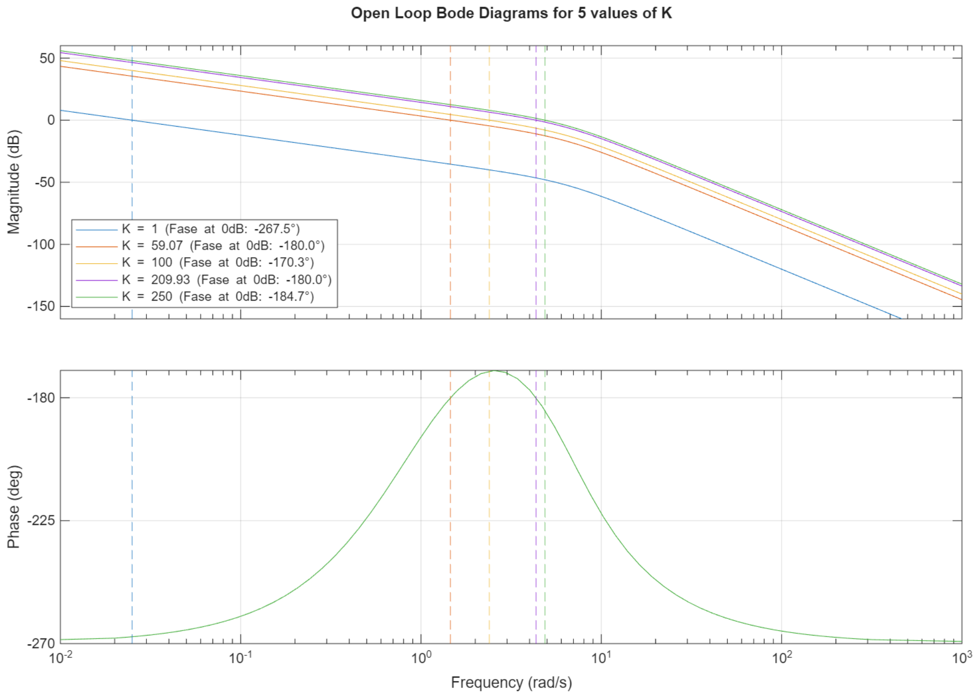

Figure 1: Open loop Bode Diagrams for $K = 1, 59.07, 100, 209.93$, and 250.

As observed in the Bode plots, the stability is strictly dependent on the controller's gain K, which shifts the magnitude curve vertically. For stable values of K (e.g., $K = 100$), the phase margin is positive because the phase is above $-180°$ ($-\pi$ rad) at the frequency where the magnitude crosses 0 dB. On the other hand, for unstable values outside the calculated range, the magnitude curve crosses the 0 dB axis when the phase has already dropped below $-180°$, resulting in a negative phase margin and, thus, an unstable closed loop response. The values of K exactly at the limits result in marginally stability.

4 Task 3: Stability Verification

To physically verify the calculated stability range and the frequency response analysis, we evaluate the system's time domain closed loop step responses.

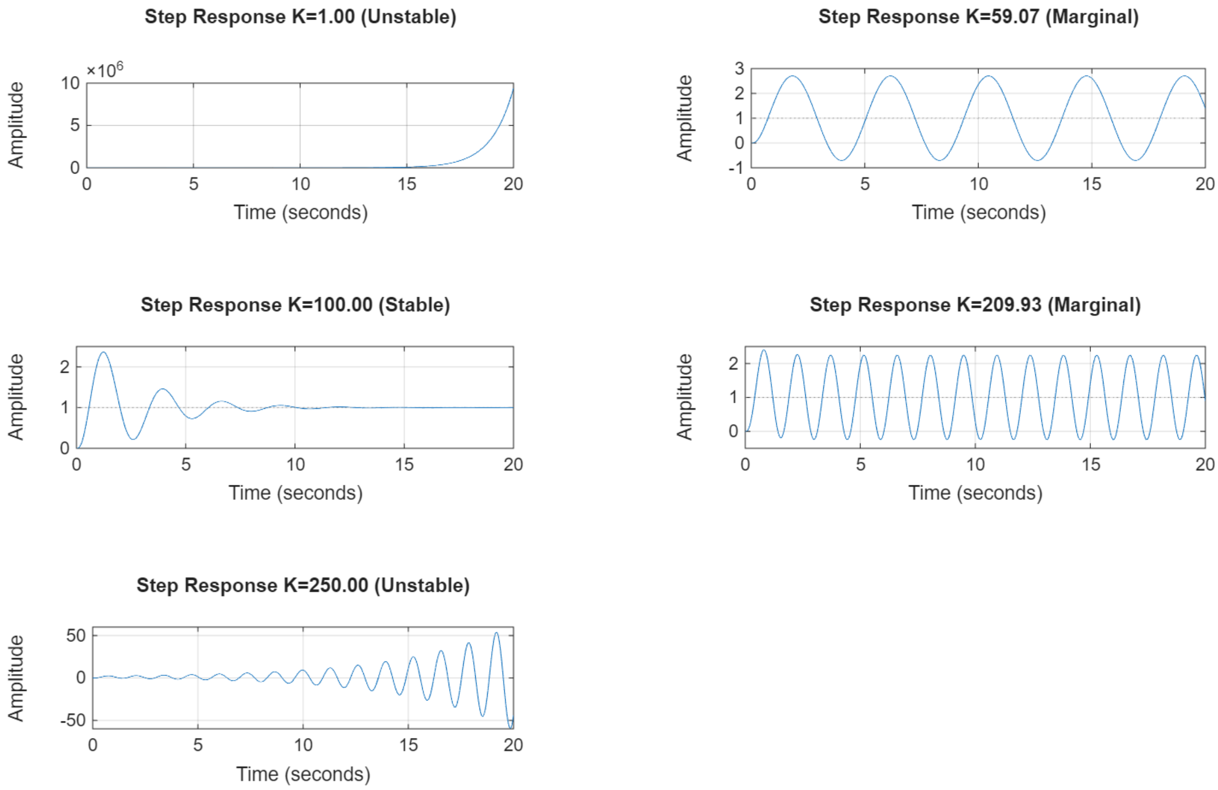

Figure 2: Closed loop step responses for $K = 1, 59.07, 100, 209.93$, and 250.

As demonstrated in Figure 2, the time-domain behavior perfectly matches the theoretical margins. The response diverges slowly for $K = 1$, achieves stabilization for $K = 100$, and exhibits severe, growing oscillations for $K = 250$. At the exact limits ($K = 59.07$ and $K = 209.93$), the system sustains constant oscillations, confirming marginally stability.

5 Conclusions

The analysis of the lateral position control system demonstrates that the human pilot must apply a very specific range of gain ($59.07 < K < 209.93$) to successfully stabilize the aircraft during a carrier landing.

Physically, this bounded stability region highlights the difficulty of the maneuver. If the pilot's response is too weak ($K < 59.07$), they fail to overcome the aircraft's inherent open loop instability (caused by the right-half plane pole at $s = 1$), leading to a divergence from the centerline. On the other hand, if the pilot is too aggressive and overcorrects ($K > 209.93$), the system's complex poles cross into the unstable region, inducing a big oscillatory instability.